Mission 2b

Now that you know how far off your camera's calibration is, you can use the drone's *hyper-photon calibrator* to have it calibrate itself. You still need to tell it how to calibrate each lens, however.

The calibrator is supposed to work by **either heating a lens or cooling it, by 1 at a time**. Testing it on the drone, however, shows that it's a bit imprecise: instead of changing the heat of only the lens you aim it at, **it also changes the heat of its immediate neighbours**. So if you heat up the second lens, the first and the third heat up too. Note that *the first lens neighbours the last*, because the lenses form a ring. Also, **the calibrator can still cause the temperatures to loop.** Each heat or cool you aim at a lens counts as one use, even though it changes three lenses.

Since every photon and every calibration just adds to a lens's state, *the order doesn't matter*: calibrating the camera before the test photo or after it gives the same result. So you can simply start from the lenses' final state and bring them all back to `0`.

Here is the first example from mission 2a. After its last frame, the lenses are:

```
2  1  3 -1  0 -2
```

Heating *lens 1* also heats *lenses 6 and 2*. Many calibrations make every lens neutral. For example, heating *lens 1* twice, cooling *lens 2* three times, heating *lens 5* once and cooling *lens 6* once works, and uses the calibrator **7 times**. This one is cheaper:

```
lens           1  2  3  4  5  6
start          2  1  3 -1  0 -2
2 heats at 1  +2 +2  .  .  . +2
3 heats at 2  +3 +3 +3  .  .  .
1 heat at 3    . +1 +1 +1  .  .
end            0  0  0  0  0  0
```

Lenses 1 to 3 are heated 7 times each, which loops them all the way round back to `0`. *Lens 6* only gets warmed through *lens 1*, its neighbour across the ring. That's **6 uses** of the calibrator, and no calibration uses fewer, so the answer for this example is **6**.

Here is the second example from mission 2a. After its last frame, the lenses are:

```
-2 -3  3  3  3  0  2  3  0
```

With 9 lenses, heating *lens 1* changes *lenses 9, 1 and 2*, and heating *lens 9* changes *lenses 8, 9 and 1*. One calibration that works is 3 heats at *lens 3*, 1 heat at *lens 4*, 3 heats at *lens 6*, 3 cools at *lens 7*, 2 cools at *lens 8* and 2 heats at *lens 9*, which uses the calibrator **14 times**. This one is cheaper:

```
lens           1  2  3  4  5  6  7  8  9
start         -2 -3  3  3  3  0  2  3  0
3 heats at 1  +3 +3  .  .  .  .  .  . +3
3 cools at 4   .  . -3 -3 -3  .  .  .  .
2 cools at 8   .  .  .  .  .  . -2 -2 -2
1 cool at 9   -1  .  .  .  .  .  . -1 -1
end            0  0  0  0  0  0  0  0  0
```

The very hot *lenses 3 to 5* are cooled straight back to `0`. *Lenses 1 and 9* both rely on the ring: *lens 1* ends at `−2 + 3 − 1 = 0`, and *lens 9* at `0 + 3 − 2 − 1 = 0`. That's **9 uses** of the calibrator, and no calibration uses fewer, so the answer for this example is **9**.

What is the smallest number of calibrator uses that calibrates your camera?